

Automated Estimation of Functional Walking Test Performance Using Video-Based Pose Estimation

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Introduction

- Walking speed is routinely used to assess functional status as well as the overall health for a range of different populations (Middleton, 2013).
- The 10MWT measures the time it takes to walk a set difference, then divides by the distance to get velocity (Shirley Ryan AbilityLab, 2014). The 2MWT measures the distance someone is able to walk within two minutes between a fixed cone distance (Willi et al., 2022).
- The 10MWT and 2MWT, which provide clinically meaningful measures of walking speed and endurance.
- Manual and in person assessments are the traditional procedure for these tests, limiting the frequency, clinical access, and increasing the burden for patients with mobility limitations.
- Recent advances in computer vision and the widespread availability of low cost video have created new opportunities for these functional assessments to be remote.

Objective

- To evaluate the whether a computer-vision pipeline can accurately capture the velocity as well as distance traveled in remote 10MWT and 2MWT video assessments.

Data Collection

- Demographics: 140 Adults ranging from healthy to functionally limited. (ages 19-100).
- Datasets: Observations were used from the CP, VA, and UVM datasets. Participants were recorded completing 10MWT and 2MWT trials.
- Ground Truth: Videos were labeled on either the time or distance traveled by trained researchers. Some videos were dual-coded to examine inter-rater reliability across research assistants, which were excellent. Labeled outcomes (which were used for model training) were compared to trained clinician/research assistant recorded outcomes at the time of the test with an ICC of 0.86-1.0

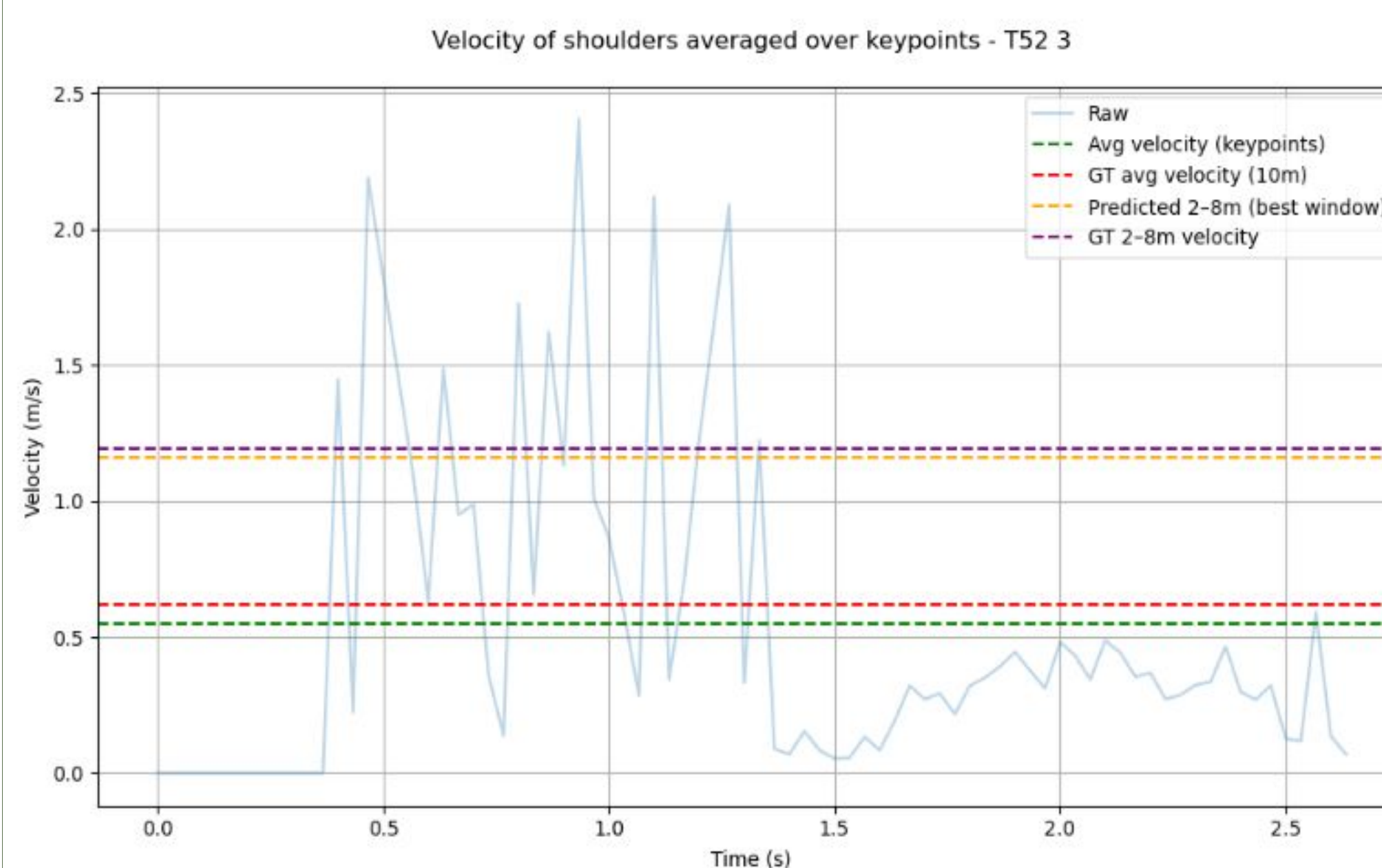
Methods

- Pose Extraction: 3D pose landmarks extracted using MediaPipe Pose (world coordinates, meters)
 - Evaluation: Compared predictions to ground truth using absolute error AE, average percent, and median percent errors for both 10m and 2-8m distances
- 10MWT**
- Extract 3D pose data - Use the shoulder midpoint
 - Smooth the trajectory - Calculate velocity
- 2MWT**
- Use shoulder oscillations to detect turns within fixed distance
 - Count cone transitions, and use avg velocity to estimate final unfinished cone stretch
 - Estimate total distance

Results

10MWT:			2MWT:	
Median error ~15–20% across tasks				
Metric	10 m Value	2-8 m Value	Metric	2 min Value
Avg Abs Error (m/s)	0.112	0.233	Avg Abs Error (ft)	95.643
Median Abs Error (m/s)	0.089	0.198	Median Abs Error (ft)	83.0
Avg % Error	19.0%	22.0%	Avg % Error	12.4%
Median % Error	15.1%	17.9%	Median % Error	16.7%

Keypoint Displacement



Shoulder displacement over time used to estimate walking velocity

Example Frame (10MWT)



Blurred video frame used for remote gait analysis

Discussion

- Our error is close to the threshold of clinically meaningful change, suggesting the method may be useful in practice. While there are a lot of changes and improvements that could be made, these results suggest that video based estimation may be capable of detecting clinically significant differences in mobility when implemented correctly.
- In order to counteract the camera angle we shifted our approach to monitor the oscillations between the two shoulder keypoints. This better reflected the structure of the test as the model explicitly counted cone to cone transitions and accumulated distance accordingly. The alignment between the model structure and the clinical test design likely explains the substantial reduction in error observed following refinement.

Conclusion & Future Work

- Automating the 10MWT and 2MWT using video based pose estimation has important implications for remote clinical care. These tests are commonly used to indicate functional status, fall risk, and rehabilitation progress.
- Future work will explore deep learning models to directly estimate gait velocity and distance from video, reducing reliance on heuristic-based methods.

References

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